

RTR-BR:

- В настоящий момент имеет:
 - **te0** - порт в сторону **ISP**
 - **te1** - порт в сторону **FW-BR**

```
ecorouter>show port brief
```

Name	Physical	Admin	Lacp	Last Change	Description
te0	UP	UP	*	19m:46s ago	
te1	UP	UP	*	19m:46s ago	

```
ecorouter>
```

- Базовая настройка:
 - имя
 - адресация в соответствие с L3
 - требуемый пользователь

```
enable
configure terminal
hostname rtr-br
ip domain-name au.team

username net_admin
password P@ssw0rd
role admin
exit

interface int0
description "Connect-to-ISP"
ip address 84.212.78.78/27
exit

port te0
service-instance te0/int0
encapsulation untagged
connect ip interface int0
exit
exit

interface int1
description "Connect-to-FW-BR"
ip address 10.2.0.1/30
exit

port te1
service-instance te1/int1
encapsulation untagged
connect ip interface int1
exit
exit

write memory
```

- Проверить:

```

rtr-br#show hostname
rtr-br
rtr-br#show run | include ip domain-name
ip domain-name au.team
rtr-br#show users localdb
User: admin
  Description: Administrator User
  Docker socket access: disabled
  VR:
    pvr
  Roles:
    admin
User: daemon
  Description: The user is used to get configuration data
  Docker socket access: disabled
  VR:
    pvr
  Roles:
    daemon
User: net_admin
  Description:
  Docker socket access: disabled
  VR:
    pvr
  Roles:
    admin
rtr-br#show ip int br

```

Interface	IP-Address	Status	VRF
int0	84.212.78.78/27	up	default
int1	10.2.0.1/30	up	default

```

rtr-br#ping 84.212.78.94
PING 84.212.78.94 (84.212.78.94) 56(84) bytes of data.
64 bytes from 84.212.78.94: icmp_seq=1 ttl=64 time=16.8 ms
64 bytes from 84.212.78.94: icmp_seq=2 ttl=64 time=15.9 ms
64 bytes from 84.212.78.94: icmp_seq=3 ttl=64 time=14.4 ms

--- 84.212.78.94 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 14.426/15.696/16.774/0.968 ms
rtr-br#

```

- Создаём интерфейс **loopback.0** и назначаем на него IP-адрес:

```
conf t
interface loopback.0
ip address 192.168.255.2/32
no shutdown
exit
```

- Проверить:

```
rtr-br(config)#do show ip interface brief loopback.0
```

Interface	IP-Address	Status	VRF
-----	-----	-----	-----
loopback.0	192.168.255.2/32	up	default

```
rtr-br(config)#
```

- Настраиваем IS-IS:

```
router isis
net 49.0001.1921.6825.5002.00
is-type level-2-only
metric-style wide
passive-interface loopback.0
exit

interface int0
ip router isis
isis circuit-type level-2-only
isis network point-to-point
exit

interface loopback.0
ip router isis
exit
```

- Проверить:

```
rtr-br(config)#do show ip route isis
IP Route Table for VRF "default"
i L2      192.168.255.1/32 [115/20] via 84.212.78.94, int0, 00:00:08

Gateway of last resort is not set
rtr-br(config)#do ping 192.168.255.1
PING 192.168.255.1 (192.168.255.1) 56(84) bytes of data.
64 bytes from 192.168.255.1: icmp_seq=1 ttl=64 time=15.3 ms
64 bytes from 192.168.255.1: icmp_seq=2 ttl=64 time=14.3 ms
64 bytes from 192.168.255.1: icmp_seq=3 ttl=64 time=13.4 ms

--- 192.168.255.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 13.410/14.324/15.286/0.766 ms
rtr-br(config)#
```

- Настраиваем BGP:

```
router bgp 64499
bgp router-id 192.168.255.2
bgp log-neighbor-changes
neighbor 192.168.255.1 remote-as 64499
neighbor 192.168.255.1 update-source loopback.0
neighbor 192.168.255.1 description iBGP-to-ISP
address-family ipv4 unicast
neighbor 192.168.255.1 activate
neighbor 192.168.255.1 next-hop-self
exit-address-family
exit

write memory
```

- Проверить:

```

rtr-br#show ip bgp summary
BGP router identifier 192.168.255.2, local AS number 64499
BGP table version is 2
1 BGP AS-PATH entries
0 BGP community entries

Neighbor      V    AS    MsgRcv   MsgSen   TblVer   InQ    OutQ    Up/Down
-----
192.168.255.1  4    64499   5         2         2        0        0      00:00:10

Total number of neighbors 1

Total number of Established sessions 1
rtr-br#

```

```

rtr-br#show ip route
Codes: C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

IP Route Table for VRF "default"
Gateway of last resort is 192.168.255.1 to network 0.0.0.0

B*      0.0.0.0/0 [200/0] via 192.168.255.1 (recursive via 84.212.78.94), 00:01:24
C       10.2.0.0/30 is directly connected, int1
C       84.212.78.64/27 is directly connected, int0
B       100.64.0.0/16 [200/0] via 192.168.255.1 (recursive via 84.212.78.94), 00:01:24
i L2    192.168.255.1/32 [115/20] via 84.212.78.94, int0, 00:01:24
C       192.168.255.2/32 is directly connected, loopback.0
rtr-br#ping 77.88.8.8
PING 77.88.8.8 (77.88.8.8) 56(84) bytes of data.
64 bytes from 77.88.8.8: icmp_seq=1 ttl=55 time=90.7 ms
64 bytes from 77.88.8.8: icmp_seq=2 ttl=55 time=89.6 ms
64 bytes from 77.88.8.8: icmp_seq=3 ttl=55 time=88.6 ms

--- 77.88.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 88.593/89.614/90.694/0.858 ms
rtr-br#

```

- Настраиваем NAT:

```
conf t

interface int0
ip nat outside
exit

interface int1
ip nat inside
exit

ip nat pool BR 10.2.0.1-10.2.2.126

ip nat source dynamic inside-to-outside pool BR overload interface int0
write memory
```

- Настраиваем OSPF в сторону FW-BR для анонсирования маршрута по умолчанию:

```
router ospf 1
passive-interface default
no passive-interface int1
network 10.2.0.0/30 area 1
default-information originate
exit

write memory
```